2.1 Metric Spaces and Topology: Problems

Attempt each problem for at least 30 minutes before reading solutions.md. Write full proofs.

Warm-up

1. For $x, y \in \mathbb{R}$ let $d'(x,y) = \dfrac{|x-y|}{1+|x-y|}$.

(a) Show that d' is a metric.
(b) Show that d' and the usual metric $d(x,y)=|x-y|$ have the same open sets.
(c) Show that $(\mathbb{R}, d')$ is bounded and complete.

2. Let X be any set with the discrete metric. Show that every subset of X is both open and closed, describe exactly which sequences converge, show X is complete, and show a subset of X is compact iff it is finite.

3. Let (X,d) be a metric space.

(a) Show that $\overline{B(x,r)} \subseteq \{y : d(x,y) \le r\}$.
(b) Give an example where the inclusion is strict.
(c) Show that equality holds in any normed vector space.

4. Let X, Y be topological spaces. Show that $f : X \to Y$ is continuous iff $f(\overline{A}) \subseteq \overline{f(A)}$ for every $A \subseteq X$.

5. Let (X,d) be a metric space and $A \subseteq X$.

(a) If X is complete and A is closed, show A is complete.
(b) If A (with the restricted metric) is complete, show A is closed in X.

6. Let X be an infinite set with the cofinite topology. Show that X is T_1 but not Hausdorff, and that every subset of X is compact.

Core

7. Show that $C[0,1]$ with the metric $d_1(f,g) = \int_0^1 |f-g|$ is not complete.

8. Show that there is exactly one continuous function $f : [0,1] \to \mathbb{R}$ satisfying

$$f(x) = e^x + \frac{1}{2}\int_0^1 \cos(xt)\, f(t)\, dt \quad \text{for all } x \in [0,1].$$

9. Show that a non-empty complete metric space with no isolated points is uncountable.

10. (a) For $f : \mathbb{R} \to \mathbb{R}$, show that the set of points where f is continuous is a G_δ set (a countable intersection of open sets).

(b) Deduce that there is no function $f : \mathbb{R} \to \mathbb{R}$ that is continuous at every rational point and discontinuous at every irrational point.

11. Let (K,d) be a compact metric space and $f : K \to K$ satisfy $d(f(x),f(y)) < d(x,y)$ whenever $x \neq y$. Show that f has a unique fixed point. Show by example that compactness cannot be replaced by completeness.

12. Work in ℓ^2 (square-summable real sequences with $\|x\| = (\sum x_n^2)^{1/2}$), which you may assume is complete.

(a) Show that the closed unit ball is closed and bounded but not compact.

(b) Show that the "Hilbert cube" $H = \{x \in \ell^2 : |x_n| \le 1/n \text{ for all } n\}$ is compact.

13. Show that $[0,1]$ is not homeomorphic to the circle S^1, and that $\mathbb{R}$ is not homeomorphic to $\mathbb{R}^2$.

14. Let $S_+ = \{(x, \sin(1/x)) : 0 < x \le 1\}$ and $S = S_+ \cup (\{0\} \times [-1,1]) \subseteq \mathbb{R}^2$. Show that S is connected but not path connected.

15. Show that a connected metric space with at least two points is uncountable.

16. Let $\mathbb{R}^{\mathbb{N}}$ denote the set of real sequences.

(a) Show that $f : \mathbb{R} \to \mathbb{R}^{\mathbb{N}}$, $f(t) = (t, t, t, \dots)$ is continuous for the product topology but not for the box topology.

(b) Let $x^{(n)}$ be the sequence whose first n terms are 0 and whose remaining terms are 1. Show that $x^{(n)} \to 0$ in the product topology but not in the box topology.

17. (a) Show that the quotient $[0,1]/(0 \sim 1)$ is homeomorphic to S^1.

(b) Show that $[0,1]^2/\sim$, where $(s,0) \sim (s,1)$ and $(0,t) \sim (1,t)$ (and the equivalence relation these generate), is homeomorphic to $S^1 \times S^1$.

18. Show that a compact subset of a Hausdorff space is closed. Give an example of a compact subset of some topological space that is not closed.

Challenge

19. Let $f : (0,\infty) \to \mathbb{R}$ be continuous, and suppose that for every $x > 0$ the sequence $f(x), f(2x), f(3x), \dots$ tends to 0. Prove that $f(x) \to 0$ as $x \to \infty$. (Hint: Baire.)

20. Let (K,d) be a compact metric space and $f : K \to K$ an isometry: $d(f(x), f(y)) = d(x,y)$ for all x, y. Show that f is surjective.

21. Prove that every compact Hausdorff space is normal.

22. Let $C = \{\sum_{n \ge 1} a_n 3^{-n} : a_n \in \{0,2\}\} \subseteq [0,1]$ (the middle-thirds Cantor set). Show that $\varphi : \{0,1\}^{\mathbb{N}} \to C$, $\varphi(x) = \sum_n 2x_n 3^{-n}$ is a homeomorphism, where $\{0,1\}^{\mathbb{N}}$ has the product of discrete topologies. You may use Tychonoff's theorem.

23. Let (X,d) be a metric space. Show that X is compact iff every continuous function $X \to \mathbb{R}$ is bounded. (You may use the Tietze extension theorem.)